

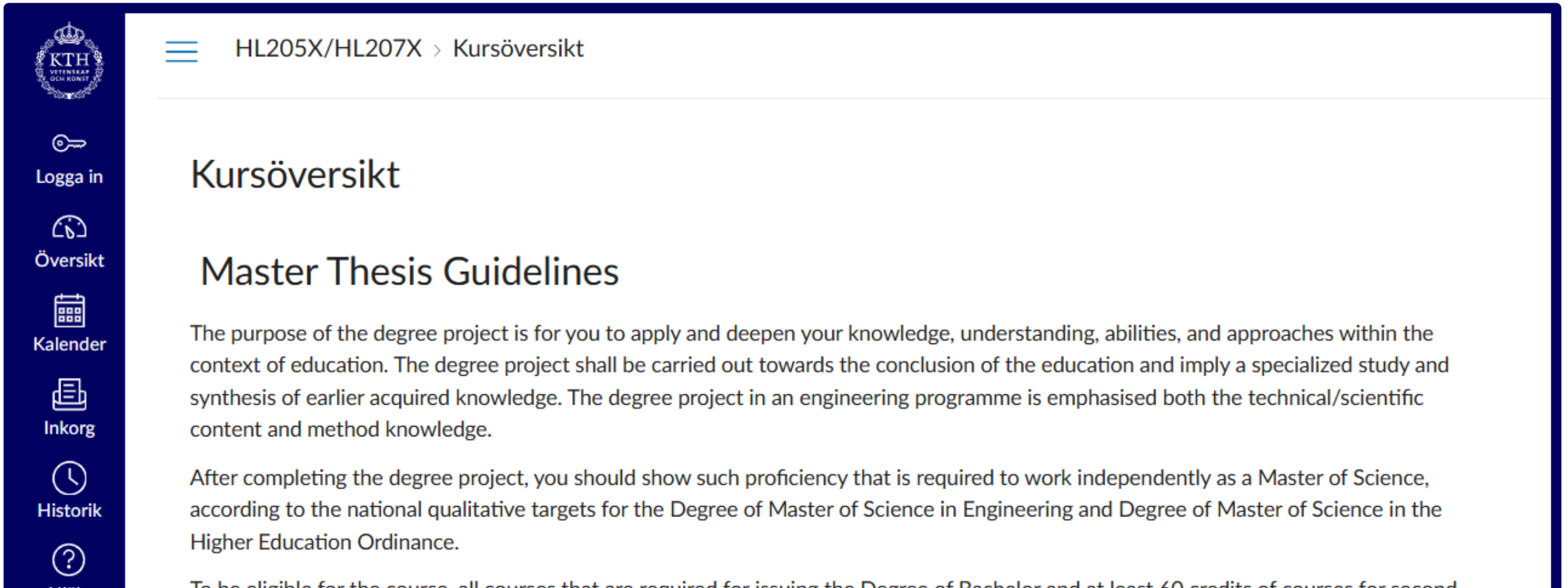
Degree Project in Medical Engineering

Intended Learning Outcomes

The purpose of the degree project is that the student shall ***apply and deepen his or her knowledge****, understanding, abilities, and approaches within the context of the education. The degree project shall be carried out towards the conclusion of the education and imply a specialised study and* ***synthesis of earlier acquired knowledge****. In the degree project in an engineering programme is emphasised both the* ***technical/scientific content and method knowledge****.*

Information About the Course

https://canvas.kth.se/courses/44148/assignments/syllabus



Find Project



- Feasible in 20 weeks full-time work
- Will fulfil degree objectives in the thesis
  - Demonstrate ability to search for, gather and integrate knowledge
  - Demonstrate ability to identify, analyse, assess and handle complex phenomena
  - …
- Work should be original, not trivial, meaningful, valid and engineering
- Try to keep it within your track (biomechanics, imaging, health systems)

Good vs bad?

Biomedical Engineering Applications in Medicine

This project will explore how biomedical engineering is applied in modern medicine. It will investigate different technologies used for diagnosis, treatment, monitoring, and rehabilitation. The project may cover medical devices, sensors, medical imaging, robotics, artificial intelligence, and other emerging technologies.

Development of an AI Ultrasound Imaging System for Early Detection of Liver Fibrosis

This project will develop and evaluate an AI-based method for detecting liver fibrosis from ultrasound images. The student will use a suitable dataset to preprocess images, train a machine-learning model, and evaluate its performance. The project will focus on developing a novel proof-of-concept system.

Improving Nurse Performance Through Motivation

This project will investigate how nurses can be motivated to improve their performance at work. It will explore nurses' experiences, opinions, and motivations through interviews. The study will identify factors that influence motivation and discuss possible strategies for improving nurse performance.

Previous Degree Projects

Artificial Intelligence and Machine Learning in Medical Imaging

Projects focused on image reconstruction, segmentation, classification, and explainable AI for diagnostic imaging



Previous Degree Projects

Biomedical Signal and Sensor Technologies

Projects focused on development or validation of wearable sensors, biosignal analysis, and signal processing for physiological monitoring



Previous Degree Projects

Biomedical Devices, Materials, and Biomechanics

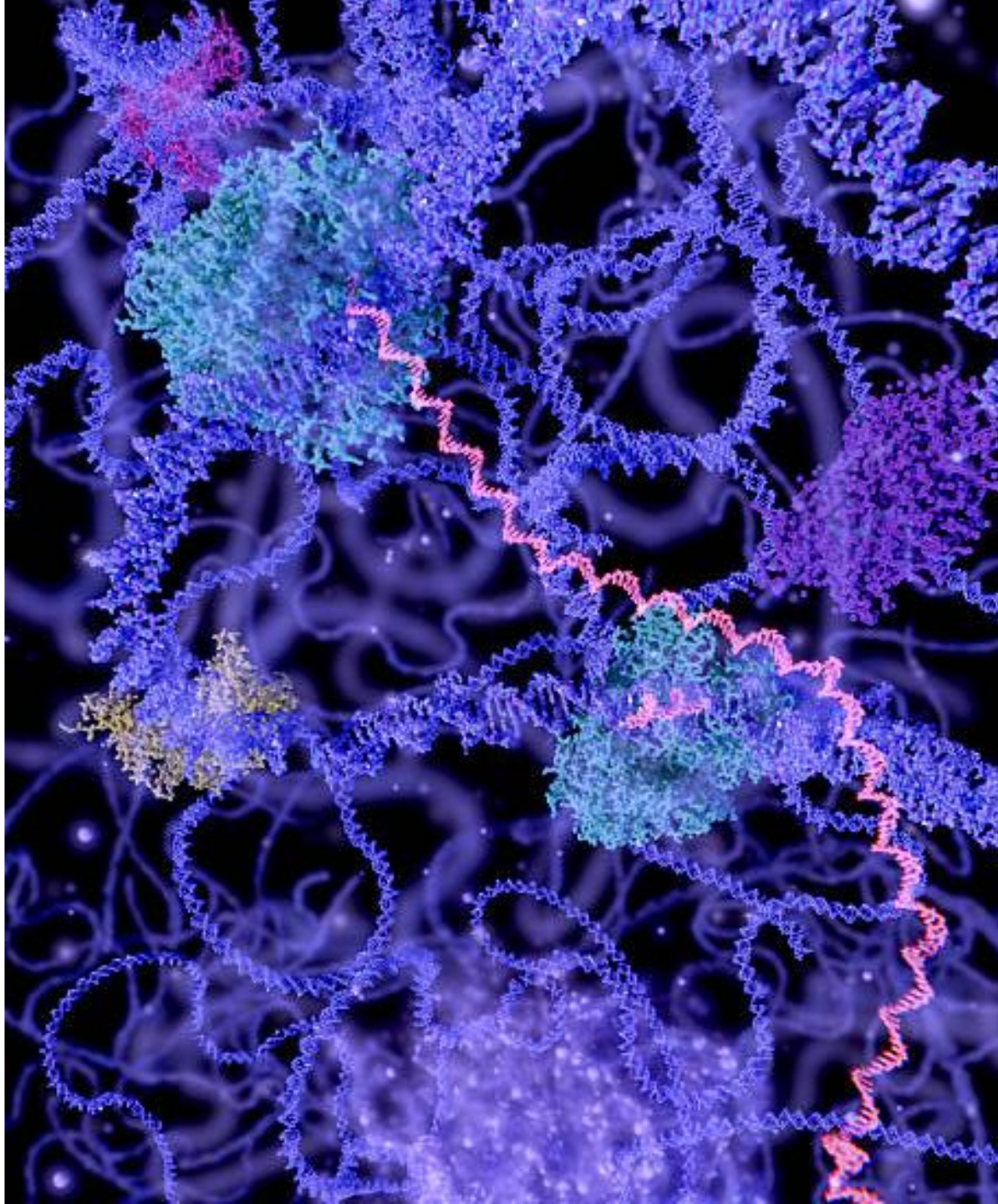
Projects focused on design and evaluation of medical devices, biomaterials, prostheses, and biomechanical models



Previous Degree Projects

Molecular, Cellular, and Nanomedical Engineering

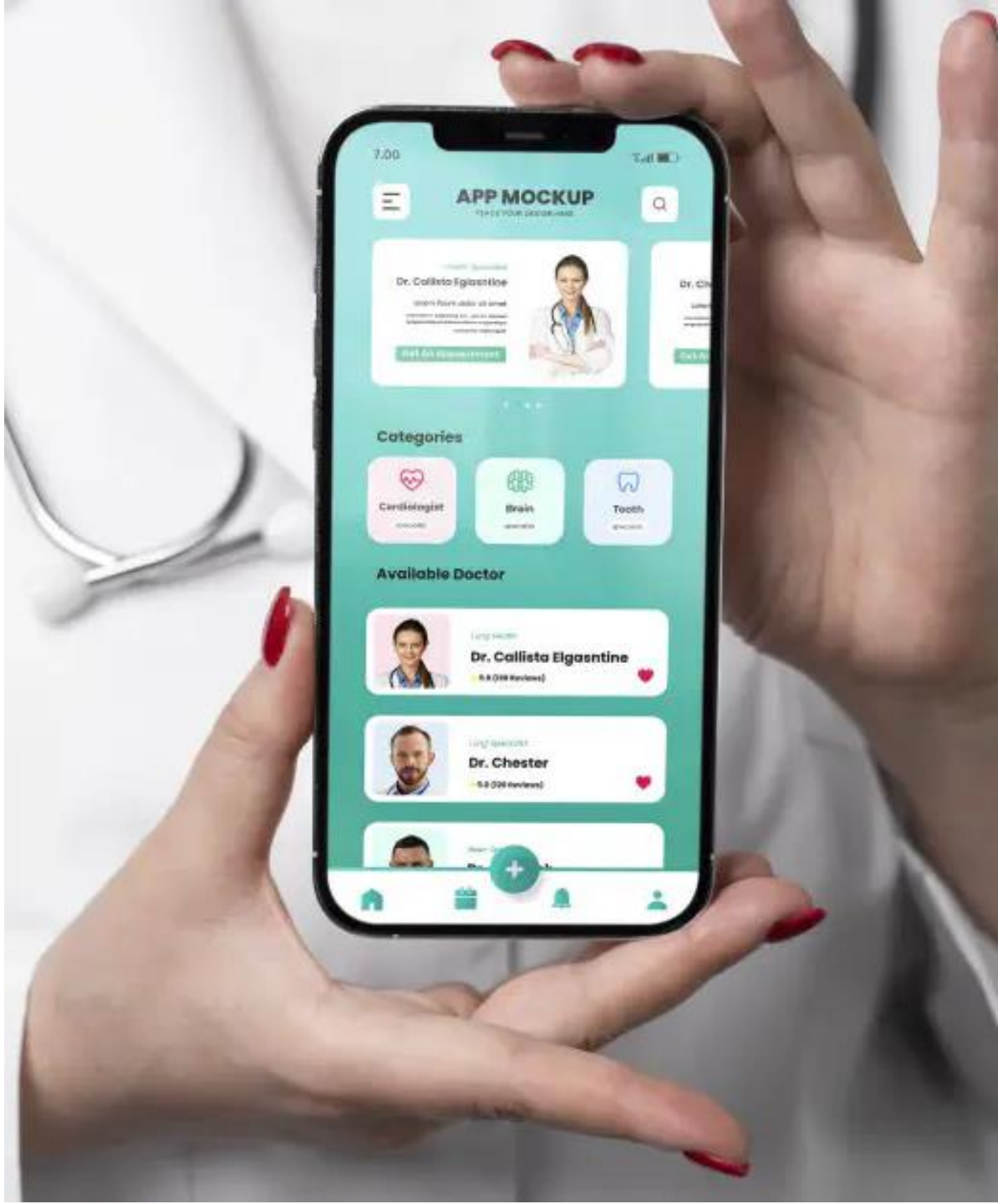
Projects focused on drug delivery systems, nanotechnology, and cell-level imaging or diagnostics



Previous Degree Projects

Digital Health, Healthcare Systems, and Decision Support

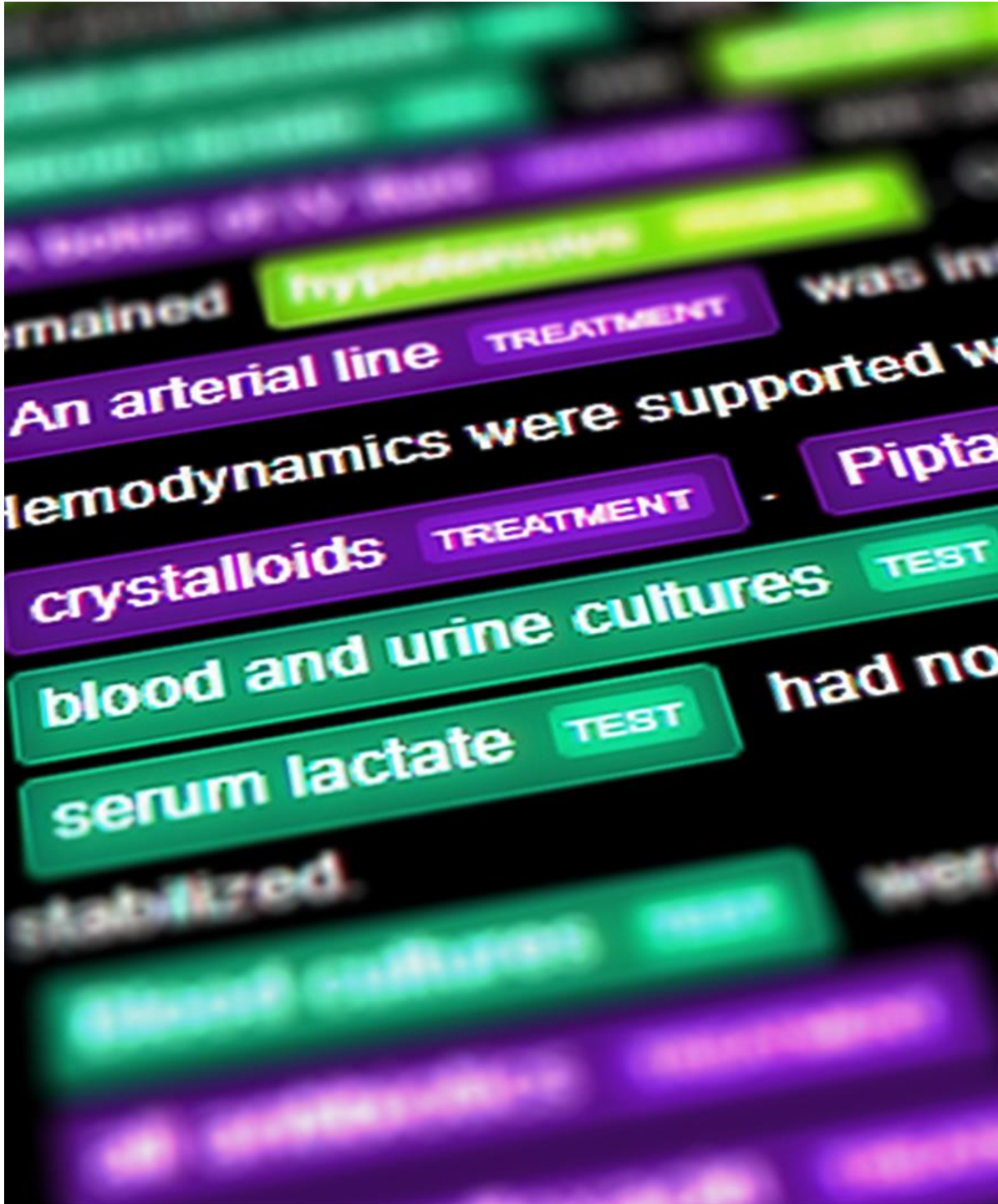
Projects focused on AI-driven or model-based healthcare optimization, digitalization, and clinical decision tools



Previous Degree Projects

Natural Language Processing and Data-Driven Medical Informatics

Projects focused on text and data mining in health records, NLP for diagnostics, and knowledge representation



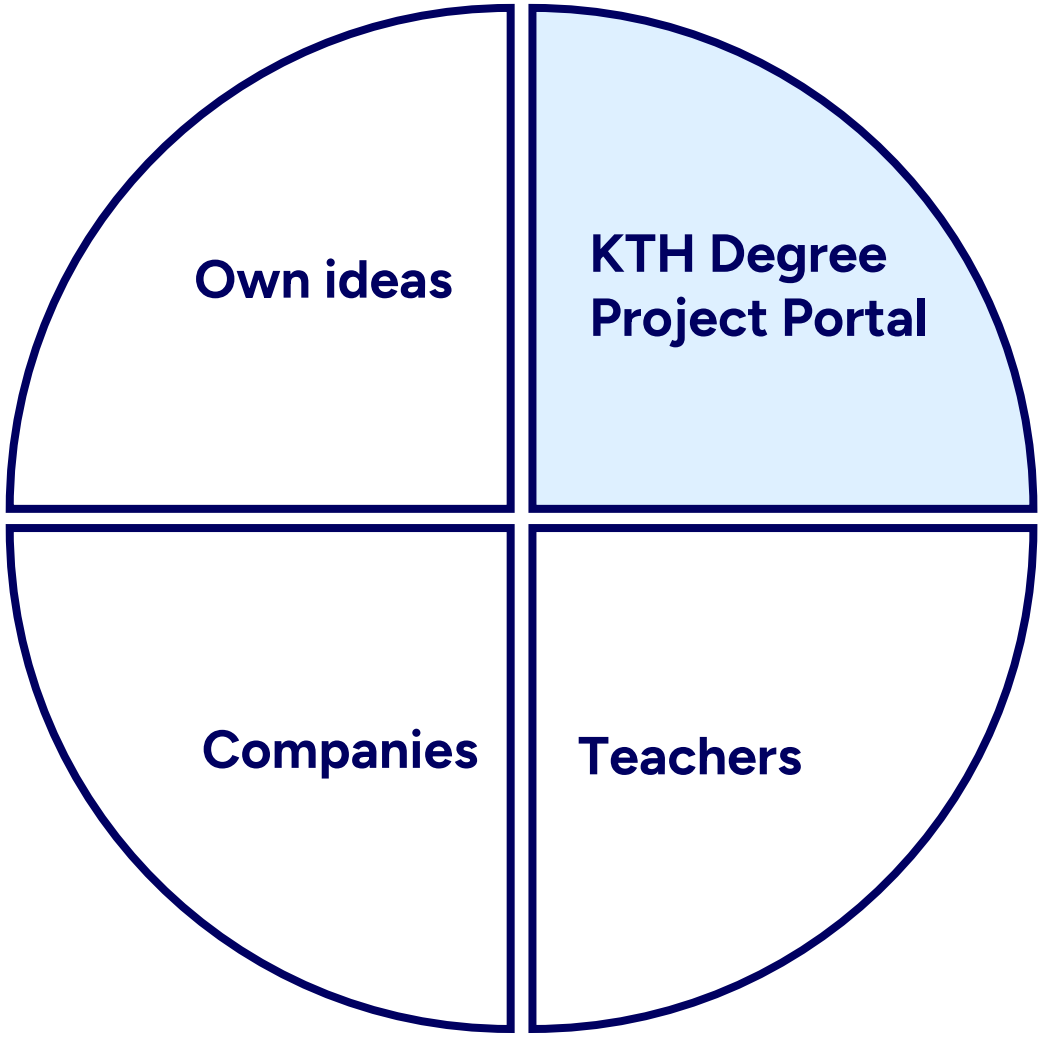
Previous Degree Projects

Human Factors, Rehabilitation, and Applied Biomechanics

Projects focused on motion capture, rehabilitation technology, ergonomics, and sport or movement analysis



Find Project



KTH Degree Project Portal

Assignments

Welcome to the KTH Degree Project Portal. Filter assignments by selecting one or more assignment types or subjects, or by using a search term.
Please observe that the KTH Degree Project Portal is not staffed between 2025-07-14 – 2025-08-03, but we will get back to you as soon as we can in the beginning of August.

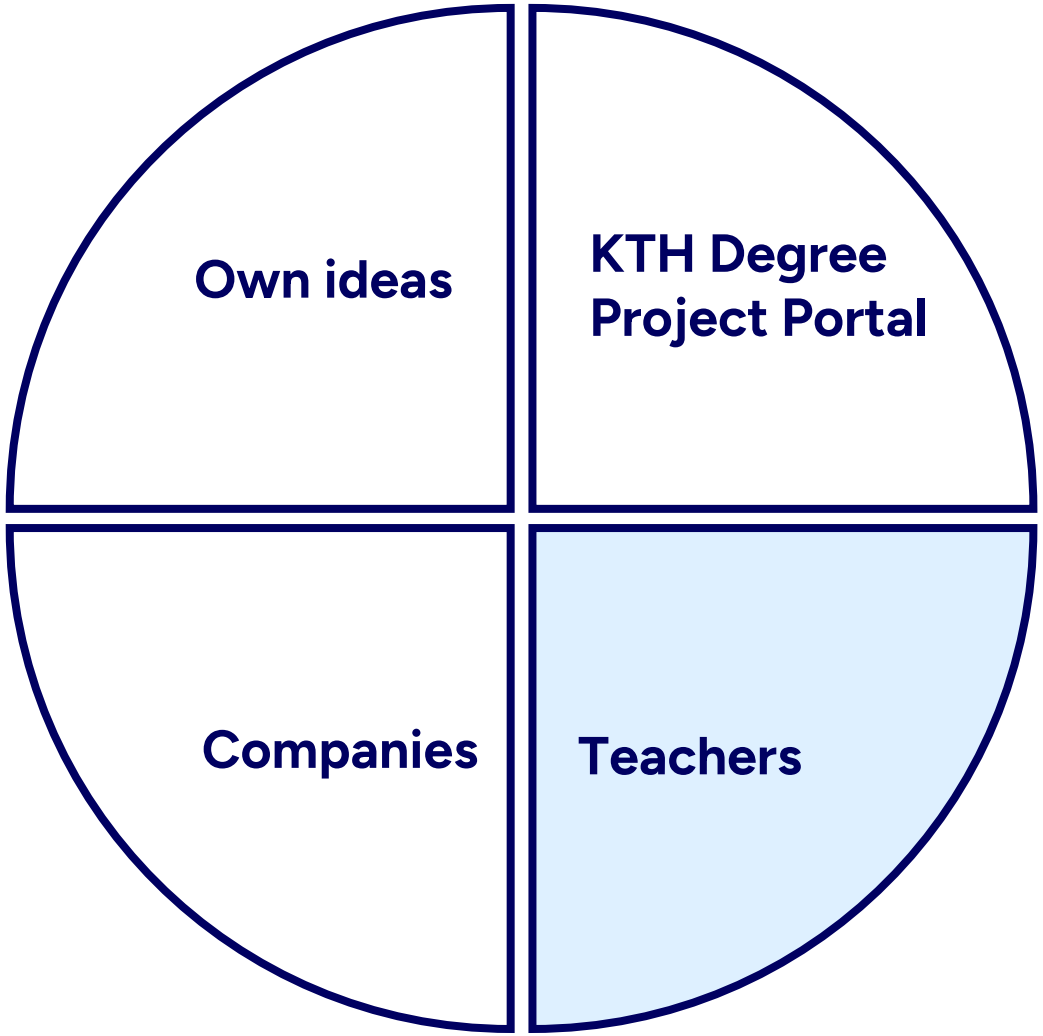
Assignment Type

- ☐ Degree project
- ☐ Seasonal and part time jobs

Subject

- ☐ Architecture and the Built Environment
- ☐ Biotechnology
- ☐ Chemical Science and Engineering

Find Project





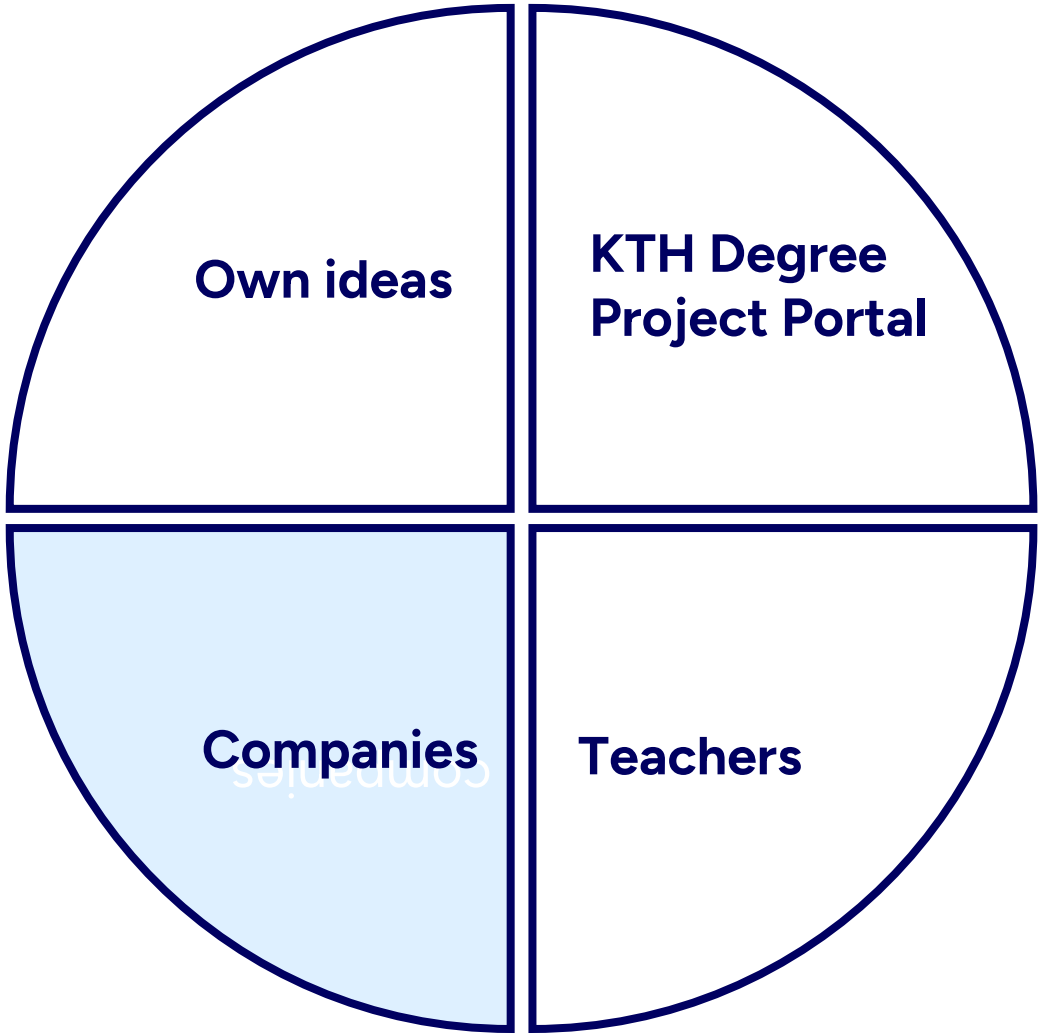








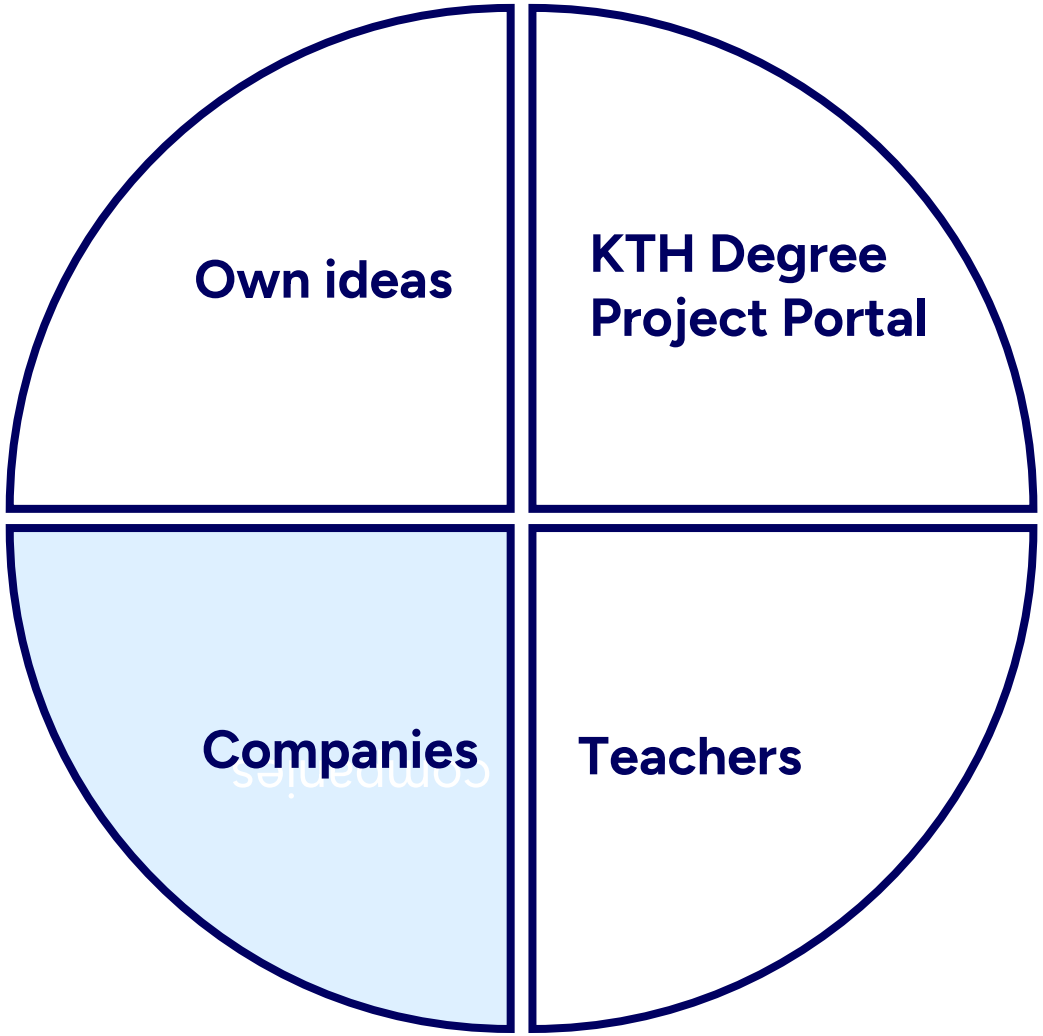
Find Project



Artificial Intelligence and Machine Learning in Medical Imaging

AI Sweden
RaySearch Laboratories
Elekta Instrument AB
Modulai AB
Digital Diabetes Analytics Sweden AB
LUMA Vision
Devyser Diagnostics
Acorai
Coala Solutions AB
SensoDetect AB
SciLifeLab
OneTwo Analytics
Doctrin AB

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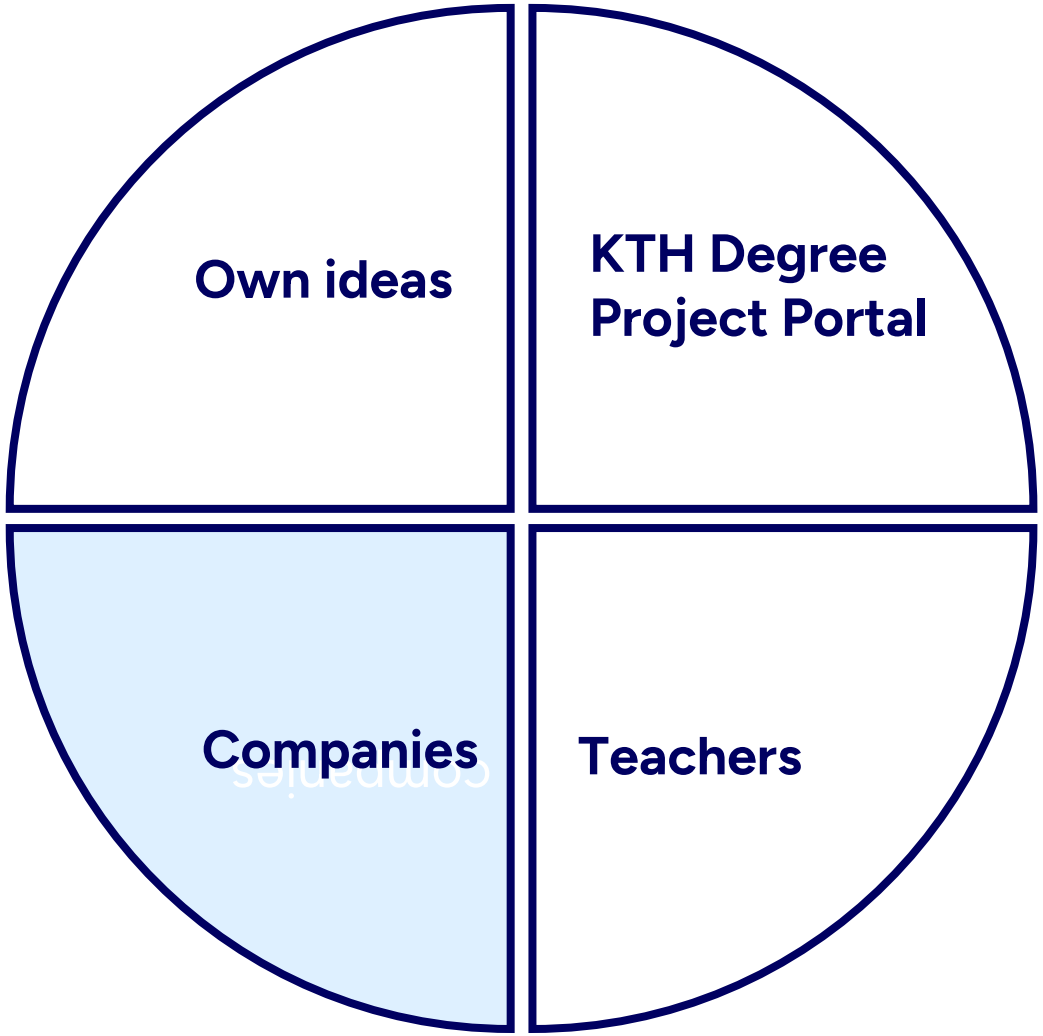
Biomedical Signal and Sensor Technologies

Coala Solutions AB
Perimed AB
Digital Diabetes Analytics Sweden AB
Acorai
SensoDetect AB
MIPS
Xsens
KTH MoveAbility Lab
Maquet Critical Care AB
Scandinavian Real Heart
Chordate Medical AB

Biomedical Devices, Materials, and Biomechanics

Maquet Critical Care AB
Scandinavian Real Heart
MIPS
Xsens
Perimed AB
Chordate Medical AB
Knightec
ÅF Digital Solutions AB (AFRY)
KTH MoveAbility Lab

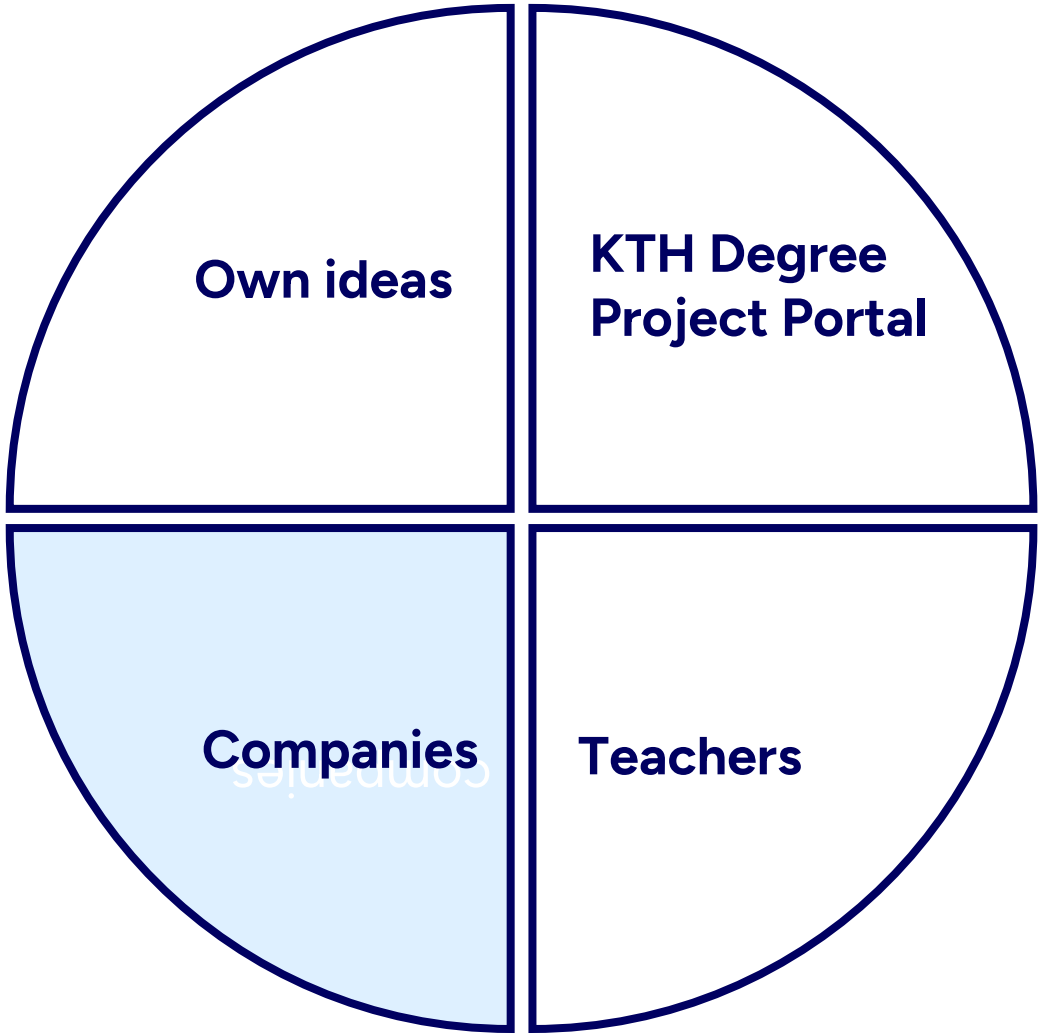
Find Project



Molecular, Cellular, and Nanomedical Engineering
Devyser Diagnostics
SciLifeLab
Karolinska Institutet

Digital Health, Healthcare Systems, and Decision Support
Doctrin AB
Atea Sverige
AI Sweden
Karolinska Institutet
Knightec
ÅF Digital Solutions AB
SciLifeLab
EverySport Group
OneTwo Analytics
Digital Diabetes Analytics Sweden AB

Find Project



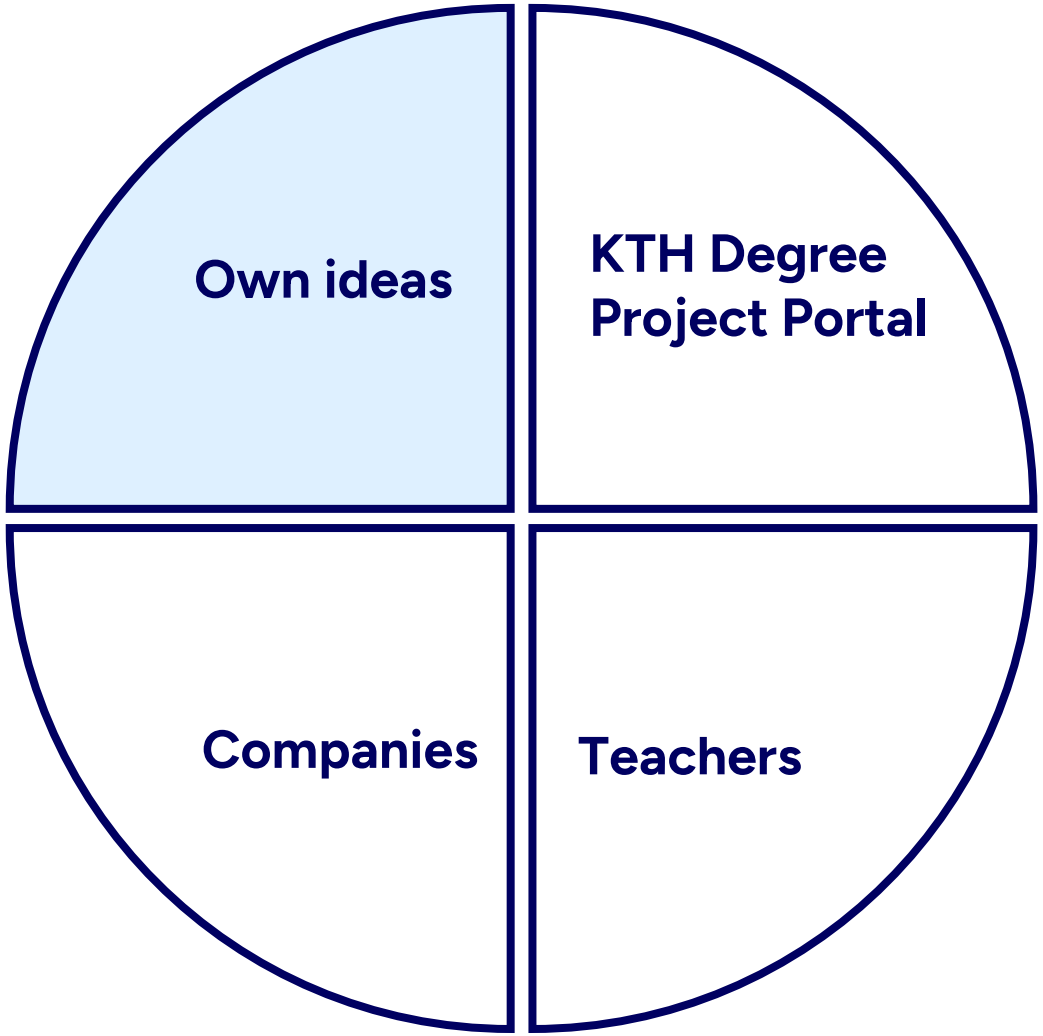
Natural Language Processing and Data-Driven Medical Informatics

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Human Factors, Rehabilitation, and Applied Biomechanics

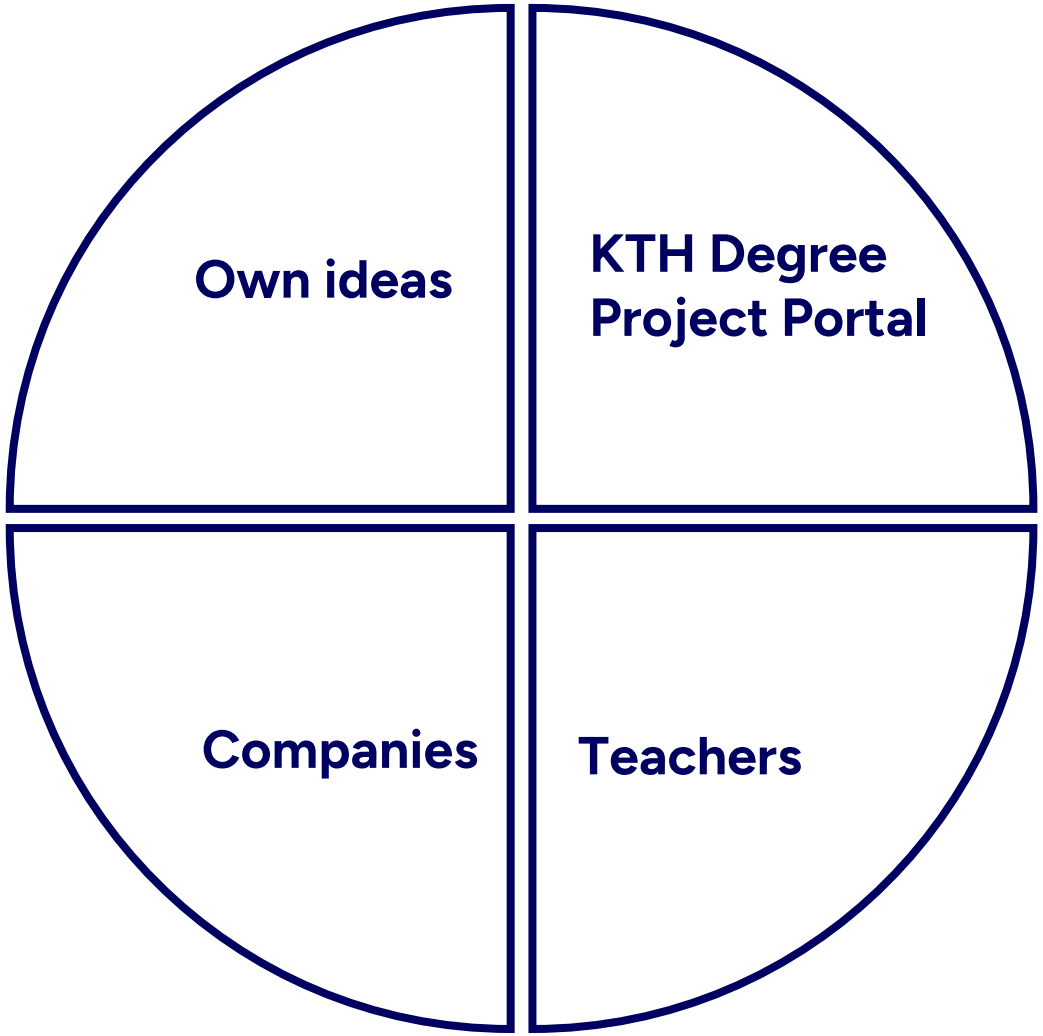
KTH MoveAbility Lab
GIH
MIPS
Xsens
Knightec
ÅF Digital Solutions AB
Spiideo
TopTracer
Live Arena Sports
Baby Journey App
EverySport Group

Find Project





Find Project



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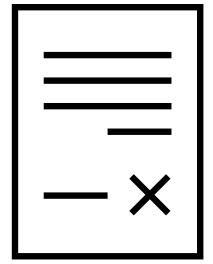
https://www.dittexjobb.se/hitta-examensarbete-exjobb

Project Proposal



- short description of the intended project (1/2 - 1 page long)
  - Project preliminary title
  - Brief description of the project
  - Project relevance to specialization within field of study
  - Supervisor
  - Confirmation that you have met course requirements
- clear current big unknown, clear goal, clear scientific methods, desirable outcomes
- submit a project proposal via the following online form: **https://www.kth.se/form/project-proposal**
- We check if we think that project has sufficiently good potential
- We provide reviewer and examiner

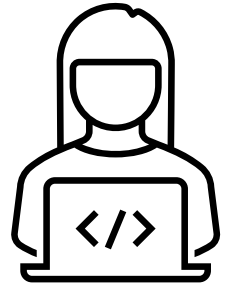
Project Description



- Project proposal is approved, you will receive project description form
- Fill it out carefully! It is your work contract
- 3- 4 pages long (reuse text later in your thesis)
- It takes time to fill it
- Once completed send it to kornevs@kth.se

- Wait for a decision
- If PASS – great, if FAILED – change or look for a new project
- most project descriptions pass
- Project description needs to be approved by **15 December**

Degree Project Process



- we register you to the course (no need to register at www.antagning.se.se/)

- Kick-off meeting
- Several meetings to help to work on your thesis
- Several reviews of other students' works
- Mid-term submission mid March
  - Rough draft of all parts
  - Showing that the proposed methodology works

- Final presentation in the beginning of June
- P/F based on feedback from reviewer and examiner

Timeline

Find Project
October - November

Project Proposal
October-November

Project Description
December 15

Degree Project Process
January - May

Final Presentation
June

